

ISBELL FIELD, AL, USA

WMO: 722111

Lat: 34.474N

Lon: 85.721W

Elev: 877

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 63847

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.4	22.4	2.8	6.6	29.6	8.6	8.8	30.7	18.9	41.4	17.3	42.7	2.7	0	0.367

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.7	93.5	73.0	91.1	72.4	89.8	72.1	77.1	87.7	76.0	85.9	74.9	84.3	5.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	131.6	81.0	73.1	126.8	80.0	72.3	123.6	79.4	41.1	87.7	40.0	85.9	38.9	84.3	81.0

Extreme Annual Design Conditions

Extreme Annual WS							Extreme Annual Temperature												n-Year Return Period Values of Extreme Temperature											
1%		2.5%	5%			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years														
Min	Max			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max													
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)															
16.3	13.1	11.6	DB	12.2	96.4	4.6	3.7	8.8	99.0	6.1	101.2	3.5	103.3	0.1	106.0															
			WB	11.1	78.3	4.6	1.9	7.8	79.7	5.1	80.8	2.5	81.8	-0.8	83.2															

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.3	40.8	44.4	51.3	59.4	67.4	75.2	77.3	77.8	72.4	61.4	50.2	44.7	
	DBStd	15.14	10.34	9.40	8.92	7.45	6.65	3.87	3.66	4.09	5.43	8.21	8.33	9.48	
	HDD50	967	321	205	94	14	0	0	0	0	0	10	98	225	
	HDD65	3266	749	579	431	195	56	0	0	0	10	168	447	631	
	CDD50	4715	37	47	134	295	540	757	845	862	672	362	104	60	
	CDD65	1541	0	1	6	27	131	308	381	397	231	55	3	1	
	CDH74	14294	0	4	81	387	1304	2963	3343	3669	2020	498	24	1	
	CDH80	5645	0	0	6	59	368	1271	1424	1646	766	104	1	0	
Wind	WSAvg	3.5	4.2	4.4	4.8	4.4	3.3	2.9	2.6	2.4	2.6	3.2	3.6	3.9	
Precipitation	PrecAvg	55.0	5.4	4.9	6.2	4.6	4.8	4.0	4.4	3.8	4.2	3.2	4.4	5.1	
	PrecMax	65.9	9.6	11.5	10.6	8.1	13.0	8.8	8.9	7.4	8.7	8.9	10.2	8.2	
	PrecMin	34.6	2.0	2.0	2.2	2.6	0.7	1.3	1.9	1.1	0.5	0.1	1.9	2.5	
	PrecStd	8.7	1.8	2.1	2.0	1.5	2.6	2.1	1.9	1.5	2.3	2.3	2.1	1.8	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.0	73.0	81.0	85.5	90.2	95.6	96.2	97.7	94.2	87.6	77.4	71.5	
		MCWB	59.3	61.6	62.6	63.8	70.8	71.7	73.6	74.6	68.7	65.8	61.4	61.6	
	2%	DB	64.4	68.5	76.6	81.6	86.6	91.6	91.8	93.7	90.7	82.1	72.4	67.9	
		MCWB	57.7	59.4	60.9	63.1	69.4	72.0	73.7	74.3	69.2	66.0	59.4	61.5	
	5%	DB	62.6	64.3	72.2	79.0	84.0	90.1	90.2	91.0	87.9	79.4	69.4	63.8	
		MCWB	56.1	54.9	58.2	61.8	68.2	71.3	73.2	73.4	70.2	64.2	58.8	58.3	
	10%	DB	58.6	61.3	68.1	75.3	81.5	87.8	87.9	88.8	84.3	76.6	64.7	60.9	
		MCWB	53.2	52.0	56.4	60.5	66.1	71.2	72.9	73.0	69.9	63.4	56.7	55.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.3	64.4	65.3	68.1	73.9	77.2	78.0	79.0	76.0	72.9	66.7	64.9
			MCDB	65.4	69.8	75.1	78.7	84.4	87.4	88.1	89.9	84.5	82.0	71.8	68.8
2%		WB	59.8	62.1	63.2	66.3	72.2	75.5	76.7	77.6	74.3	70.7	64.3	62.9	
		MCDB	64.6	66.9	71.7	76.0	82.0	86.1	86.5	88.8	82.9	77.6	68.7	65.8	
5%		WB	57.2	58.4	61.1	64.7	70.6	74.3	75.7	76.4	72.9	68.8	61.8	59.9	
		MCDB	61.6	63.4	69.0	74.1	79.3	84.6	84.9	86.7	81.5	75.1	66.2	63.2	
10%		WB	53.6	54.5	58.9	63.1	68.9	73.2	74.6	75.1	71.6	66.5	58.6	55.9	
		MCDB	56.9	58.7	65.9	72.2	77.0	83.0	83.4	84.6	79.8	73.3	63.8	60.3	
Mean Daily Temperature Range		5% DB	MDBR	19.3	20.8	22.9	25.4	23.3	22.2	20.0	20.7	22.0	23.5	23.1	18.7
			MCDBR	22.7	25.3	28.0	28.8	26.6	26.5	23.3	24.1	26.9	26.6	27.3	21.0
	MCWBR		16.2	16.1	14.6	12.9	10.9	9.5	8.0	7.9	9.5	12.1	15.6	15.5	
	5% WB	MCDBR	18.1	20.3	22.4	23.5	21.4	21.7	19.9	21.2	19.8	18.9	18.3	17.4	
		MCWBR	14.7	15.4	13.1	12.0	9.8	9.2	8.0	7.6	8.5	10.8	14.0	15.1	
Clear-Sky Solar Irradiance	taub	0.316	0.320	0.359	0.394	0.447	0.483	0.535	0.532	0.426	0.374	0.338	0.316		
	taud	2.505	2.489	2.406	2.295	2.189	2.125	1.993	1.990	2.278	2.409	2.455	2.523		
	Ebn at Noon	284	295	290	283	268	257	243	241	264	271	272	277		
	Edh at Noon	27	30	35	42	47	49	56	56	40	32	28	25		
All-Sky Solar Radiation	RadAvg	793	958	1316	1708	1876	1969	1869	1741	1500	1234	897	673		
	RadStd	57	124	121	112	147	151	121	94	139	157	102	67		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (40 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+259	+189

Nomenclature: See separate page